

Autonomous Warehouse Inventory Audit Robot (WAREOps)

An autonomous mobile robot that patrols warehouse aisles, inspects shelves with computer vision, and continuously verifies inventory — closing the gap between what the system records and what is physically on the shelf.

LiDAR Mapping | Autonomous Navigation | Camera-Gimbal Inspection

TEAM MEMBERS

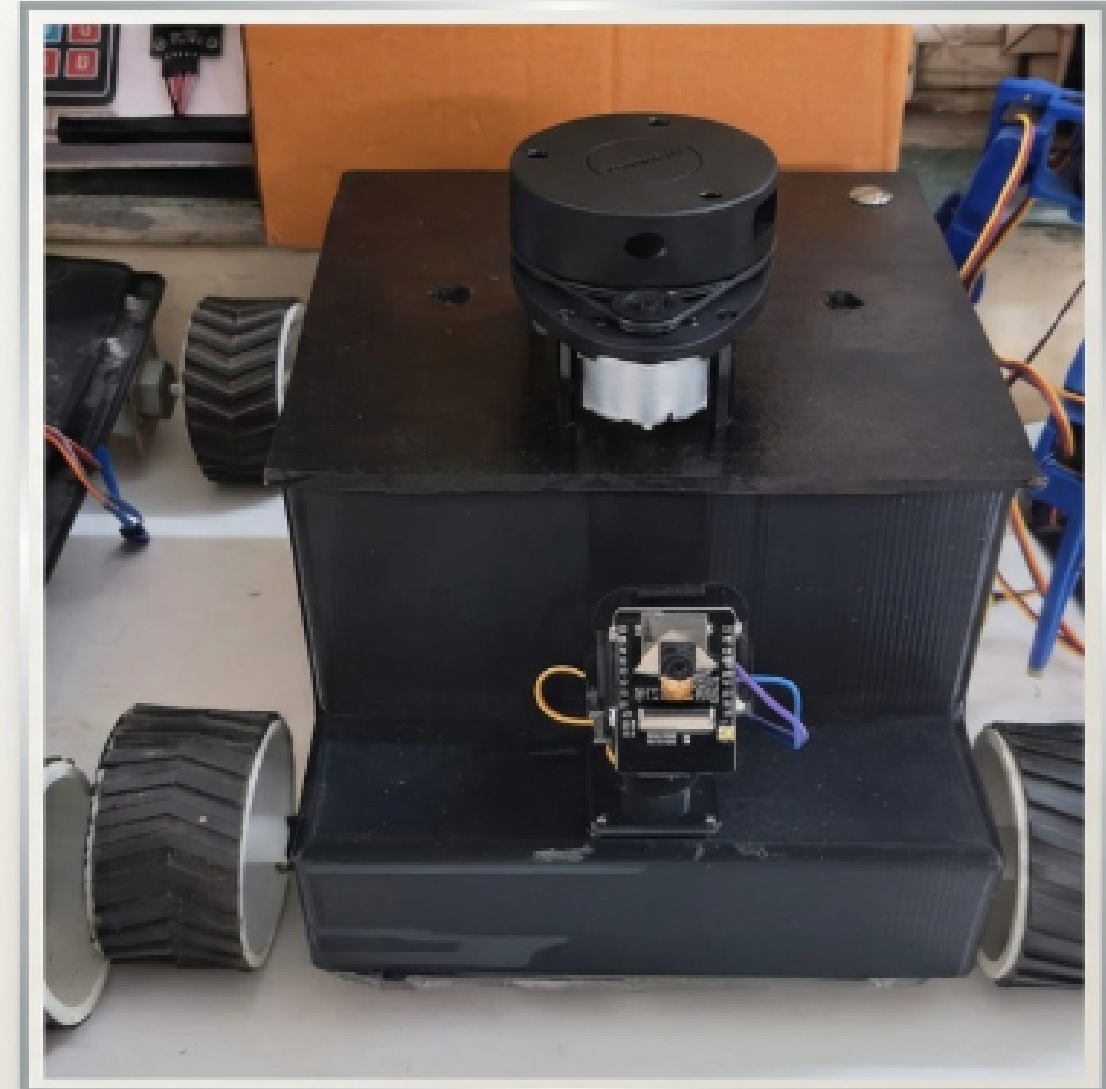
Abhinav Goel • Aanshu Kashyap • Alok Kumar Mishra • Utkarsh Kumar • Anshuman Singh

PROJECT GUIDE

Prof. Prabin Kumar Padhy

DEPARTMENT / INSTITUTION

Electronics & Communication Engineering



Warehousing Is Entering a New Era

Rapid growth in e-commerce and omnichannel retail has redefined warehouse operations. Fulfillment centers now move millions of items daily across thousands of SKUs, while delivery windows have compressed from days to hours.



Warehouses are becoming increasingly automated — but automation alone does not guarantee inventory visibility.

Industry Leaders Are Investing at Scale

Warehouse automation has become one of the fastest-growing categories in industrial technology.

WAREHOUSE AUTOMATION MARKET

\$29.98B



→ \$65.74B by 2031

- 13.98% CAGR

WAREHOUSE ROBOTICS

16.8%

Projected CAGR through 2034

- Fastest-growing warehouse technology segment

INDUSTRY LOGOS

amazon

≈1M robots deployed

walmart

\$520M robotics program

Symbotic partnership targets 400 pickup & delivery centers.

At scale, more inventory is moving through automated systems than ever before — raising the stakes for accuracy.

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The Cost of Poor Inventory Visibility

Why should we care about inventory accuracy?

FINANCIAL IMPACT

\$90–100B+ / year

U.S. retail inventory shrink from theft, operational errors, damage, and inaccuracies.

Walmart example ≈\$681B annual revenue. A 1% inventory loss ≈ **\$6.8B** potential impact.

OPERATIONAL IMPACT

- Lost sales
- Delayed shipments
- Longer search times
- Manual rework
- Reduced productivity
- Customer dissatisfaction

\$90–100B+

RETAIL SHRINK

~65%

INVENTORY RECORD ERRORS

~1 in 6

CUSTOMER REQUESTS AFFECTED

100s / week

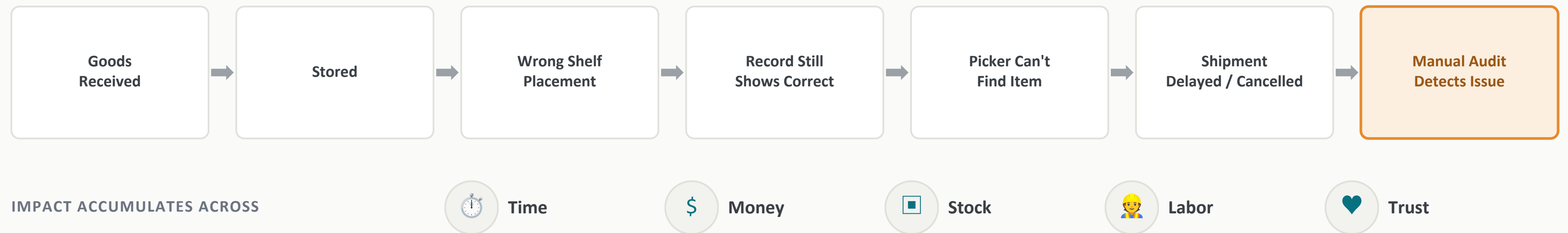
LABOR HOURS LOST

Inventory inaccuracies cost far more than the missing product — they disrupt the entire supply chain.

Figures reflect U.S. retail field studies; used for directional context.

Why Does Inventory Become Invisible?

How do these losses actually occur?

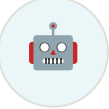


A misplaced product is often physically present — but operationally invisible.

Downstream cost compounds silently: what starts as a shelf error ends as a cancelled order and an audit finding it too late.

Why Existing Approaches Still Leave a Gap

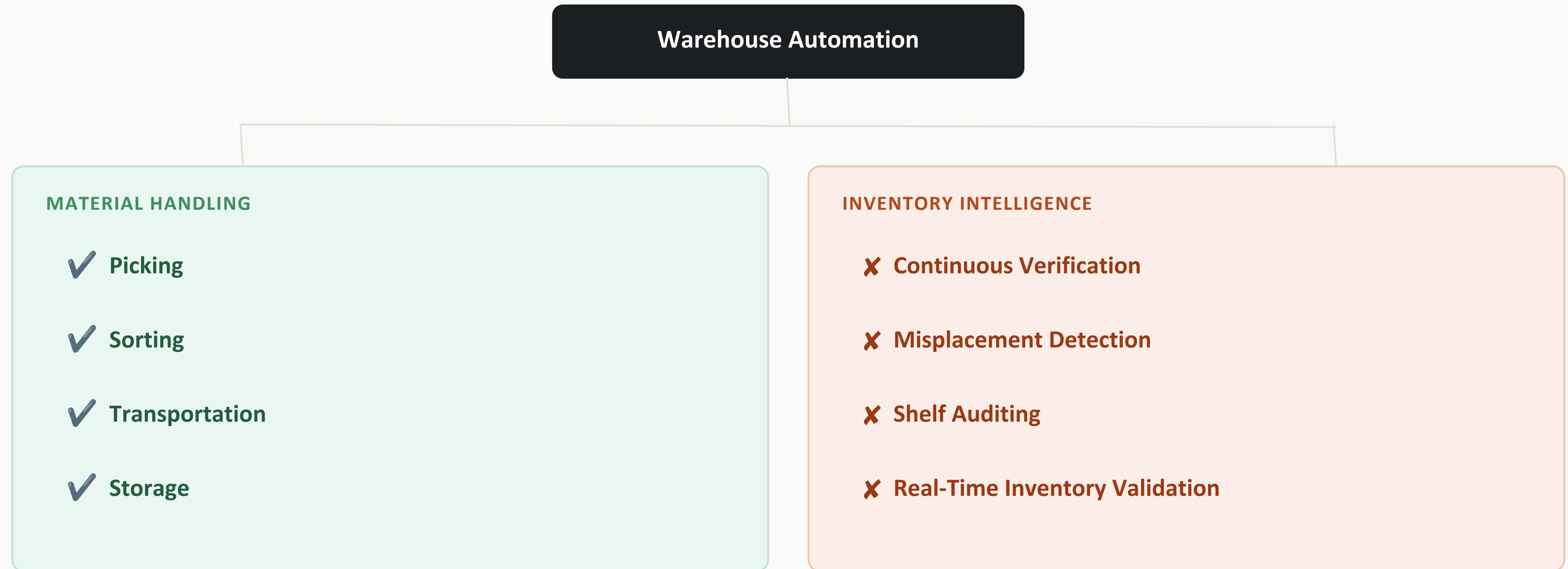
If companies invest so much, why does this still happen?

EXISTING APPROACH	PRIMARY STRENGTH	WHY MISPLACEMENT PERSISTS
 Manual Audits	Low setup cost	Periodic, labor-intensive, not real-time
 Barcode / QR	Accurate identification	Requires manual scanning
 RFID	Faster tracking	High infrastructure and tag cost
 Fixed Cameras	Continuous observation	Blind spots, occlusion, limited viewpoints
 Warehouse Robots	Material movement	Optimized for transport, not continuous auditing

Today's automation focuses on:  **Transporting**  **Picking**  **Sorting**
...rather than:  **Continuously verifying correct storage location**

The challenge is no longer moving inventory — it is maintaining continuous visibility of inventory.

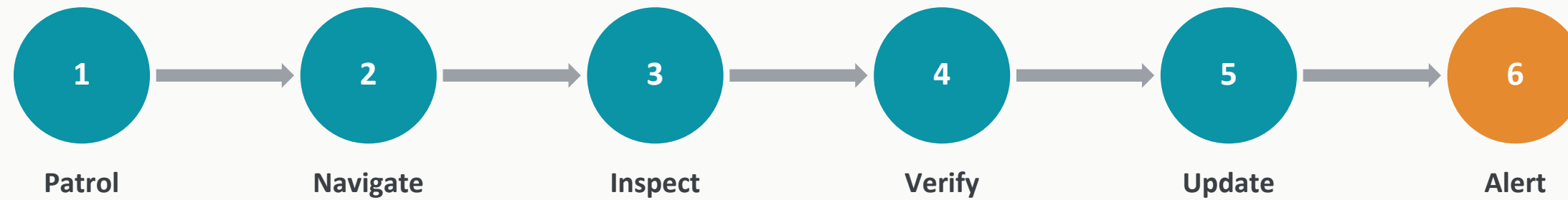
The Missing Layer in Warehouse Automation



Current automation moves products efficiently. Inventory verification still depends on manual inspections and reactive discovery.

This is the operational gap our project addresses.

Our Autonomous Inventory Auditing Robot



- Navigates warehouse aisles (ROS2 + SLAM)
- Inspects shelves using computer vision
- Reads QR codes at each location
- Detects misplaced inventory
- Updates inventory records automatically
- Alerts operators in real time
- Supports FIFO / FEFO compliance



No warehouse redesign. No dense camera network. No RFID infrastructure. Just continuous inventory visibility.

Expected Operational Impact

What changes for the warehouse after deployment.

BEFORE	AFTER
Periodic manual audits	Continuous autonomous auditing
Inventory searched manually	Inventory verified continuously
Delayed discrepancy detection	Real-time alerts
Reactive corrections	Preventive monitoring
Higher labor dependency	Reduced manual effort
Limited inventory visibility	Continuous inventory intelligence

Our solution complements existing warehouse automation — adding the missing layer of continuous inventory intelligence, so warehouses know not only where products should be, but where they actually are.